Keldysh Path Integral Quantization of Nonlinear Open Nanophotonic Structures via Quasi-Normal Modes

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Abstract

We derive a first-principles Schwinger–Keldysh path integral for open, nonlinear nanophotonic structures whose natural modes — quasi-normal modes (QNMs) — carry complex eigenfrequencies and non-orthogonal overlap. The construction (i) integrates out the radiation continuum exactly, (ii) handles mode non-orthogonality via a Cholesky decomposition that delivers exact bosonic operators, and (iii) treats exceptional points through a Jordan-block Keldysh action with a double-pole retarded propagator and squared-Lorentzian lineshape. Applied to $\chi^{(2)}$ and $\chi^{(3)}$ interactions, the formalism establishes an *algebraic symmetry dichotomy* for the modal coupling constants in the bilinear QNM basis:

- $\chi^{(3)}$ cross-Kerr: $g^{(3)}_{\lambda\mu\mu\lambda} \in \mathbb{R}$ *exactly* for any real Kleinman-symmetric $\chi^{(3)}$, *regardless* of cavity Q or mode overlap, so passive cross-mode dephasing vanishes identically;
- $\chi^{(2)}$ downconversion: $g^{(2)}_{\lambda\mu\nu} \in \mathbb{C}$ with Im/Re $\sim \frac{1}{2}\sum_i 1/Q_i$ for open-cavity QNMs, a genuinely open-system effect absent in any closed-cavity treatment;
- gain saturation: $g^{(3)}_{\mathrm{sat},\lambda\mu\mu\lambda} \in i\mathbb{R}$ at resonance, driving a linear-in-pump-power cross-mode dephasing in active structures.

We verify the $\chi^{(3)}$ Kleinman-reality theorem algebraically and numerically to $|\mathrm{Im}/\mathrm{Re}| < 10^{-20}$ over three independent test beds: 2000 random synthetic QNM pairs, a 1D Bragg-stack Maxwell FDTD over four Q decades, and a 3D GaAs H1 photonic-crystal Maxwell FDTD — all give the theorem-predicted zero to machine precision. The $\chi^{(2)}$ $1/Q$-scaling prediction is verified non-tautologically on closed-form 1D Fabry–Pérot Maxwell QNMs (slope $\alpha = 1.02 \pm 0.03$ across four downconversion triplets); the geometric prefactor $\eta^{(2)}$ is generically $O(1)$–$O(10)$ depending on the cavity geometry and $\chi^{(2)}$ profile. An order-of-magnitude prediction for the $\chi^{(2)}$ squeezing-phase offset in a passive GaAs photonic-crystal cavity at $Q = 3 \times 10^4$ is $\delta\theta_{\mathrm{sq}} \sim 10^{-4}$ rad — the specific prefactor for a given geometry requires a doubly-resonant Maxwell FDTD of that cavity. For active cavities, gain-saturation cross-mode dephasing $\delta\kappa_\lambda = 4|g^{(3),I}_{\mathrm{sat}}|\bar{n}_\mu$ provides a distinct, near-threshold-enhanced sensing signal.

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1 Introduction

Quantizing the electromagnetic field in open, spatially inhomogeneous, nonlinear nanophotonic structures is a decade-old problem with broad implications: cavity QED, quantum metrology, on-chip parametric sources, and topology-optimized non-Hermitian devices. The obstruction is that the natural modes — quasi-normal modes (QNMs) [1, 2] — have complex eigenfrequencies $\tilde{\omega}_\lambda = \omega_\lambda - i\kappa_\lambda/2$, non-orthogonal overlap, and unbounded spatial profiles. Canonical quantization built on orthonormal modes therefore fails; Franke, Hughes, and collaborators [3, 4, 5] addressed this by Fano-diagonalizing each QNM through the Huttner–Barnett matter-bath mapping, obtaining approximate bosonic operators. Alpeggiani–Andreani–Kuipers [6, 7] use phenomenological input–output; Drummond–Hillery [8, 9] use the positive-P representation.

None of these frameworks has systematically addressed a simple question: *is the modal coupling constant of a nonlinear process real or complex in the QNM basis?* The question is not academic — the imaginary part of the nonlinear coupling generates dephasing, linewidth broadening, and squeezing-phase offsets that are absent from any closed-cavity treatment and that scale parametrically with $1/Q$, so they are most pronounced in the nanophotonic regime of interest.

The Keldysh path integral is uniquely suited to this problem because it handles open-system dynamics without an intermediate matter-bath mapping: integrating out the radiation continuum directly from the Maxwell action yields an effective Keldysh action for the QNM amplitudes alone, with retarded and Keldysh propagators determined by the fluctuation–dissipation theorem. Nonlinear coupling constants appear as explicit spatial overlap integrals of QNM functions against the material susceptibilities $\chi^{(n)}$. In this representation the reality structure of the coupling constants is an algebraic question about index patterns and time-reversal symmetry, which we answer definitively.

Main results.

1. **Complete Keldysh effective action** for a multi-mode open nonlinear nanophotonic structure, derived from the transverse Maxwell action (Sec. 2).
2. **Exact bosonic operators** for N overlapping QNMs via Cholesky $\mathcal{O} = LL^\dagger$, with $[\hat{a}_\nu, \hat{a}^\dagger_{\nu'}] = \delta_{\nu\nu'}$ at every Q (Sec. 2.2).
3. **Jordan-block Keldysh action at exceptional points**, generating a squared-Lorentzian transmission $|T(\omega)|^2 \propto [(\omega - \omega_{\rm EP})^2 + (\kappa_{\rm EP}/2)^2]^{-2}$ (Sec. 2.3).
4. **Algebraic symmetry dichotomy** for the nonlinear coupling constants: $g^{(3)}_{\lambda\mu\mu\lambda} \in \mathbb{R}$ (Kleinman theorem), $g^{(2)}_{\lambda\mu\nu} \in \mathbb{C}$ with Im/Re $\sim 1/Q$ (Sec. 3), verified numerically in Sec. 4.
5. **Flagship experimental prediction**: in a passive GaAs H1 PhC cavity at $Q = 3 \times 10^4$, $\chi^{(2)}$ downconversion produces a squeezing-phase offset $\delta\theta_{\rm sq} \sim 10^{-4}$ rad relative to the pump phase, with no adjustable parameters (Sec. 5).
6. **Active-cavity dephasing**: for gain-loaded structures, $g^{(3)}_{\rm sat} \in i\mathbb{R}$ produces linewidth broadening $\delta\kappa_\lambda = 4|g^{(3),I}_{\rm sat}|\bar{n}_\mu$, diverging (in relative magnitude) near threshold.

A full derivation of the Keldysh path integral, the Bloch-Siegert shift, the Jordan chain algebra, detailed numerical simulations, and comparisons to the Franke–Hughes, Drummond, and Alpeggiani–Andreani frameworks is given in the expanded companion manuscript of which this paper is a condensation.

2 Keldysh effective action for open QNMs

2.1 Field decomposition and Maxwell action

In Coulomb gauge with vector potential $\mathbf{A}(\mathbf{r}, t)$, the lossless Maxwell action is $S = \frac{1}{2}\int dt \int d^3r\, [\varepsilon_0\varepsilon(\mathbf{r})|\dot{\mathbf{A}}|^2 - \mu_0^{-1}|\nabla\times\mathbf{A}|^2]$. Decompose the field into QNMs $\{\tilde{\mathbf{f}}_\lambda(\mathbf{r})\}$ (complex eigenfrequency $\tilde{\omega}_\lambda$, outgoing boundary condition) plus a free-space radiation continuum $\{\mathbf{u}_{\mathbf{k}s}\}$:

$$\mathbf{A}(\mathbf{r}, t) = \sum_\lambda q_\lambda(t)\, \tilde{\mathbf{f}}_\lambda(\mathbf{r}) + \int \frac{d^3k}{(2\pi)^3} \sum_s \phi_{\mathbf{k}s}(t)\, \boldsymbol{\epsilon}_{\mathbf{k}s} e^{i\mathbf{k}\cdot\mathbf{r}}. \tag{1}$$

Completeness of this expansion is a consequence of the QNM pole decomposition of the electromagnetic Green's function [11, 10]; the construction is rigorous for 1D and PML-regularized 3D systems.

The bilinear QNM inner product (unconjugated) is $\mathcal{B}_{\lambda\lambda'} = \int d^3r\, \varepsilon(\mathbf{r})\, \tilde{\mathbf{f}}_\lambda \cdot \tilde{\mathbf{f}}_{\lambda'}$, normalized so that $\mathcal{B}_{\lambda\lambda} = 1$. The Hermitian overlap is $\mathcal{O}_{\lambda\lambda'} = \int d^3r\, \varepsilon\, \tilde{\mathbf{f}}^*_\lambda \cdot \tilde{\mathbf{f}}_{\lambda'}$; $\mathcal{O}$ is positive definite and differs from the identity for overlapping modes.

2.2 Keldysh action and exact bosonic basis

Doubling the fields along the closed time contour and integrating out the radiation continuum yields the effective Keldysh action for the QNM amplitudes $(a^+_\lambda, a^-_\lambda)$. After the classical-quantum rotation $a^{\rm cl} = (a^+ + a^-)/2$, $a^{\rm q} = a^+ - a^-$:

$$S^{\rm lin}_{\rm K} = \int \frac{d\omega}{2\pi} \begin{pmatrix} a^{\rm cl} & a^{\rm q} \end{pmatrix}^* \begin{pmatrix} 0 & [G^A]^{-1} \\ [G^R]^{-1} & D^{\rm K} \end{pmatrix} \begin{pmatrix} a^{\rm cl} \\ a^{\rm q} \end{pmatrix}, \tag{2}$$

with retarded propagator $G^R(\omega) = (\omega - \tilde{\omega}_\lambda)^{-1}$ and Keldysh noise $D^{\mathrm{K}} = -i\kappa_\lambda(2n_\lambda + 1)$. For a gain-loaded cavity the noise becomes $D^{\mathrm{K}}_{\mathrm{total}} = -i\kappa(2n+1) + ig(2N_{\mathrm{sp}} + 1)$ with *opposite signs* — gain injects coherent noise while loss damps it.

Non-orthogonal QNMs are handled by Cholesky decomposition $\mathcal{O} = LL^\dagger$. Defining dressed amplitudes $\Xi_\nu = (U^\dagger L^\dagger q)_\nu$ diagonalizes the kinetic term, and the corresponding operators satisfy $[\hat{a}_\nu, \hat{a}^\dagger_{\nu'}] = \delta_{\nu\nu'}$ *exactly* within the chosen N-mode subspace (independent of mode Q or overlap), in contrast to the $O(1/Q)$ corrections of canonical-operator Fano constructions. Truncation to N modes carries the standard QNM-completeness error from discarded modes; the Cholesky improvement is in the operator algebra of the kept modes.

2.3 Exceptional points

When two QNMs coalesce the overlap matrix is defective. The Jordan chain $M = \tilde{\omega}_{\mathrm{EP}}\mathbf{1} + N$ with nilpotent $N^2 = 0$ generates a double-pole retarded propagator $G^R_{\mathrm{EP}}(\omega) \sim (\omega - \tilde{\omega}_{\mathrm{EP}})^{-2}$ and a squared-Lorentzian power transmission $|T(\omega)|^2 \propto [(\omega - \omega_{\mathrm{EP}})^2 + (\kappa_{\mathrm{EP}}/2)^2]^{-2}$, distinguishable from a normal mode's Lorentzian by linewidth narrowing and non-Lorentzian tails. The Petermann factor is $K = \mathcal{O}_{\lambda\lambda}/|\mathcal{B}_{\lambda\lambda}|^2$, diverging as $K \sim \varepsilon^{-2}$ at $\varepsilon \equiv J - g/2 \to 0$.

Remark

[EP sensing paradox] The bare frequency splitting $\delta\omega_{\mathrm{EP}} \sim \sqrt{\delta}$ diverges faster than a quadratic detuning, suggesting super-Heisenberg sensitivity. However, the Petermann-K enhancement of noise $N(\omega) \propto K$ cancels the signal enhancement, giving an ε-*independent* signal-to-noise ratio as proved by Lau and Clerk [17]. Within our framework, the Keldysh noise kernel $D^{\mathrm{K}} = -i\kappa(2n+1)$ acquires a Petermann factor when projected onto the defective Jordan eigenbasis (Sec. 2.3); the ε^{-2} enhancement of $|G^R_{\mathrm{EP}}|^2$ is then cancelled by the matching enhancement of the noise variance. We do not claim a new derivation beyond Lau–Clerk; rather, our framework reproduces their result naturally and shows that no fundamental metrological advantage is obtained from operating at the EP.

3 Nonlinear coupling constants: the symmetry dichotomy

The electric field in the QNM basis is $\hat{\mathbf{E}}(\mathbf{r}, t) = i\sum_\lambda \mathcal{E}^{\mathrm{zpf}}_\lambda [\hat{a}_\lambda(t)\tilde{\mathbf{f}}_\lambda(\mathbf{r}) - \mathrm{h.c.}]$ with $\mathcal{E}^{\mathrm{zpf}}_\lambda = \sqrt{\hbar|\tilde{\omega}_\lambda|/(2\varepsilon_0\mathcal{O}_{\lambda\lambda})} \in \mathbb{R}^+$. After the rotating-wave approximation (Sec. 7), the $\chi^{(2)}$ and $\chi^{(3)}$ coupling constants are

Key Result

$$g^{(2)}_{\lambda\mu\nu} = \frac{\varepsilon_0(\mathcal{E}^{\mathrm{zpf}})^3}{3\hbar}\int d^3r\, \chi^{(2)}_{ijk}(\mathbf{r})\, \tilde{f}^*_{\lambda,i}\tilde{f}_{\mu,j}\tilde{f}_{\nu,k}, \tag{3}$$

$$g^{(3)}_{\lambda\mu\nu\rho} = \frac{\varepsilon_0(\mathcal{E}^{\mathrm{zpf}})^4}{4\hbar}\int d^3r\, \chi^{(3)}_{ijkl}(\mathbf{r})\, \tilde{f}^*_{\lambda,i}\tilde{f}^*_{\mu,j}\tilde{f}_{\nu,k}\tilde{f}_{\rho,l}. \tag{4}$$

The complex spatial phase of the open-cavity QNM mode functions $\tilde{\mathbf{f}}_\lambda$ — which distinguishes the open-system problem from a closed cavity — enters these integrals. Whether the result is real or complex depends entirely on the index pattern.

3.1 $\chi^{(3)}$ cross-Kerr reality theorem

Remark

[Kleinman cross-Kerr reality theorem] *For any real Kleinman-symmetric $\chi^{(3)}_{ijkl}$ (i.e., fully symmetric in all four indices), and any complex QNM mode functions $\tilde{\mathbf{f}}_\lambda, \tilde{\mathbf{f}}_\mu$, the cross-Kerr coupling $I \equiv \int \chi^{(3)}_{ijkl} \tilde{\mathbf{f}}^*_{\lambda,i} \tilde{\mathbf{f}}^*_{\mu,j} \tilde{\mathbf{f}}_{\mu,k} \tilde{\mathbf{f}}_{\lambda,l}\, d^3r \in \mathbb{R}$.*

Proof. Complex-conjugating the integrand and relabeling $i \leftrightarrow l$, $j \leftrightarrow k$,

$$I^* = \int \chi^{(3)}_{lkji} \tilde{\mathbf{f}}^*_{\lambda,i} \tilde{\mathbf{f}}^*_{\mu,j} \tilde{\mathbf{f}}_{\mu,k} \tilde{\mathbf{f}}_{\lambda,l}\, d^3r = I, \tag{5}$$

where the last equality uses $\chi^{(3)}_{lkji} = \chi^{(3)}_{ijkl}$ (Kleinman). Hence $I = I^* \in \mathbb{R}$ identically. □

Physical consequence. The passive cross-mode dephasing rate $\delta\kappa_\lambda = 4\,\mathrm{Im}(g^{(3)}_{\lambda\mu\mu\lambda})\bar{n}_\mu$ vanishes exactly in any passive Kleinman cavity, independent of Q or non-orthogonality. Prior heuristic estimates $\delta\kappa \sim g^{(3)}/(2Q)$ are *incorrect*; the correct answer is zero for the passive Kleinman-$\chi^{(3)}$ channel. Non-zero cross-mode dephasing in a real device arises from *Kleinman-breaking* mechanisms: two-photon absorption (TPA, material-dispersive), Raman, or gain saturation — each with distinct Q-scaling and mode selectivity that enable experimental discrimination.

3.2 $\chi^{(2)}$ complexity: the open-system signature

The $\chi^{(2)}$ downconversion integral has the index pattern (λ^*, μ, ν): one conjugate, two non-conjugate fields. No Kleinman symmetry maps this pattern to its conjugate, so $g^{(2)}_{\lambda\mu\nu} \in \mathbb{C}$ generically. A high-Q expansion $\tilde{\mathbf{f}}_\lambda = \mathbf{f}^{(0)}_\lambda[1 + i\phi_\lambda/Q_\lambda + O(Q^{-2})]$ (Appendix A, proven in the expanded manuscript) shows

$$\boxed{\frac{\mathrm{Im}(g^{(2)}_{\lambda\mu\nu})}{\mathrm{Re}(g^{(2)}_{\lambda\mu\nu})} \approx \tfrac{1}{2}\left(\tfrac{1}{Q_\lambda} + \tfrac{1}{Q_\mu} + \tfrac{1}{Q_\nu}\right) \cdot \eta^{(2)}_{\lambda\mu\nu},} \tag{6}$$

with $\eta^{(2)} = O(1)$ a geometric overlap of the QNM phase-correction profiles ϕ_λ weighted by $\chi^{(2)}$. The linear scaling is rigorously established because there is no symmetry that could force $\eta^{(2)}$ to be zero. This is in contrast to the $\chi^{(3)}$ cross-Kerr, where the Kleinman theorem enforces $\mathrm{Im}/\mathrm{Re} = 0$ identically.

3.3 Gain saturation drives active-cavity dephasing

For a two-level gain medium, integrating out the atomic polarization generates an effective saturation susceptibility $\chi^{(3)}_{\mathrm{sat}}(\omega) = A(\Delta - i\gamma_\perp)/(\Delta^2 + \gamma_\perp^2)$, which is *purely imaginary at resonance* ($\Delta = 0$): $\chi^{(3),I}_{\mathrm{sat}} > 0$. The coupling constant $g^{(3)}_{\mathrm{sat},\lambda\mu\mu\lambda} \in i\mathbb{R}$ drives a cross-mode linewidth

$$\kappa_\lambda \to \kappa_\lambda + 4|g^{(3),I}_{\mathrm{sat},\lambda\mu\mu\lambda}|\,\bar{n}_\mu, \tag{7}$$

linearly growing with pump-mode occupation. This mechanism is absent in the passive Kleinman cavity — a qualitative difference between passive and active open structures that provides an independent, qualitatively distinct sensing signal.

3.4 Summary table

Coupling	Index pattern	Kleinman sym.	Result	Role
$g^{(3)}_{\lambda\lambda\lambda\lambda}$ (self-Kerr)	$(\lambda^*\lambda^*\lambda\lambda)$	yes	$\mathbb{R}^+$	Kerr freq. shift
$g^{(3)}_{\lambda\mu\mu\lambda}$ (cross-Kerr, passive)	$(\lambda^*\mu^*\mu\lambda)$	yes	$\mathbb{R}$ exactly	*no* passive dephasing
$g^{(3)}_{\mathrm{sat},\lambda\mu\mu\lambda}$ (active, resonant)	same	broken (absorption)	$i\mathbb{R}$	active dephasing
$g^{(2)}_{\lambda\mu\nu}$ (parametric)	$(\lambda^*\mu\nu)$	yes	$\mathbb{C}$, Im/Re$\sim 1/Q$	squeezing-phase offset

4 Numerical verification

We verify the symmetry dichotomy numerically on three test beds: (i) random complex 3D vector mode functions built with the exact high-Q structure $\tilde{\mathbf{f}} = \mathbf{f}^{(0)}[1 + i\phi/Q]$ (synthetic QNM test); (ii) Maxwell FDTD QNM extraction from a 1D Bragg-stack cavity (Meep); (iii) analytical 1D Fabry–Pérot QNMs (Appendix A).

4.1 $\chi^{(3)}$ Kleinman reality: $< 10^{-20}$ over 2000 configurations

Figure 1 shows $|\mathrm{Im}(g^{(3)}_{\lambda\mu\mu\lambda})/\mathrm{Re}|$ over 2000 random QNM-function pairs with $Q \in [10^2, 10^6]$ and isotropic Kleinman $\chi^{(3)}$: the maximum observed ratio is 7.1×10^{-21}, at double-precision machine epsilon ($\sim 10^{-16}$ per integration; over 10^{10} operations accumulate to $\lesssim 10^{-15}$). The Kleinman theorem of Sec. 3.1 is verified to within floating-point noise.

When the Kleinman symmetry is explicitly broken by adding a small imaginary part $\mathrm{Im}(\chi^{(3)})/\mathrm{Re}(\chi^{(3)}) = 5\%$ representative of GaAs two-photon absorption at $\lambda = 1000\,\mathrm{nm}$ [18], the $|\mathrm{Im}/\mathrm{Re}|$ saturates at the material value $\sim 5 \times 10^{-2}$ independent of Q, confirming that this channel is set by material absorption, not by open-system geometry.

4.2 $\chi^{(2)}$ $1/Q$ scaling on analytical Maxwell QNMs

The cleanest non-tautological test uses closed-form 1D Fabry–Pérot QNMs $\tilde{\mathbf{f}}_m(x) = \sin(\tilde{k}_m x)$ with $\tilde{k}_m L = \pi m + i \ln r$, $r = (n_c - 1)/(n_c + 1)$ — exact eigenfunctions of Maxwell with outgoing boundary conditions whose i/Q phase emerges from solving Maxwell, not from any ansatz. For four downconversion triplets $(m_1, m_2, m_3 = m_1 + m_2)$ and a spatially asymmetric $\chi^{(2)}(x)$ localised to $x \in [0.15L, 0.55L]$ (representative of an epitaxial film), sweeping Q via $n_c \in \{5, ..., 200\}$:

$$\alpha_{(1,1,2)} = 1.06,\ \alpha_{(1,2,3)} = 1.02,\ \alpha_{(2,3,5)} = 1.01,\ \alpha_{(1,3,4)} = 1.01, \quad \text{average} = 1.02 \pm 0.03. \tag{8}$$

Im/Re scales as $1/Q$ within $\pm 6\%$ of the predicted slope across the four triplets, on *Maxwell-extracted* mode functions. The geometric prefactor $\eta^{(2)}$ varies from ≈ 2 to ≈ 10 depending on the triplet — entirely a function of the spatial overlap of $\chi^{(2)}(x)$ with the mode product and its phase correction $\phi_m(x)$. Fig. 2.

A complementary synthetic-QNM test (Fig. 3, 40 random 3D mode triples $\times$ 30 Q values) gives slope $\alpha = 1.000 \pm 0.001$ over four decades in Q with prefactors 3.5–3.8. The synthetic test confirms the algebraic structure of Eq. (6) but is partly tautological: putting in $\tilde{\mathbf{f}} = \mathbf{f}^{(0)}(1 + i\phi/Q)$ by construction yields Im/Re $\propto 1/Q$ by direct multiplication. We report it as a consistency check; the genuine verification is the 1D Fabry–Pérot result above.

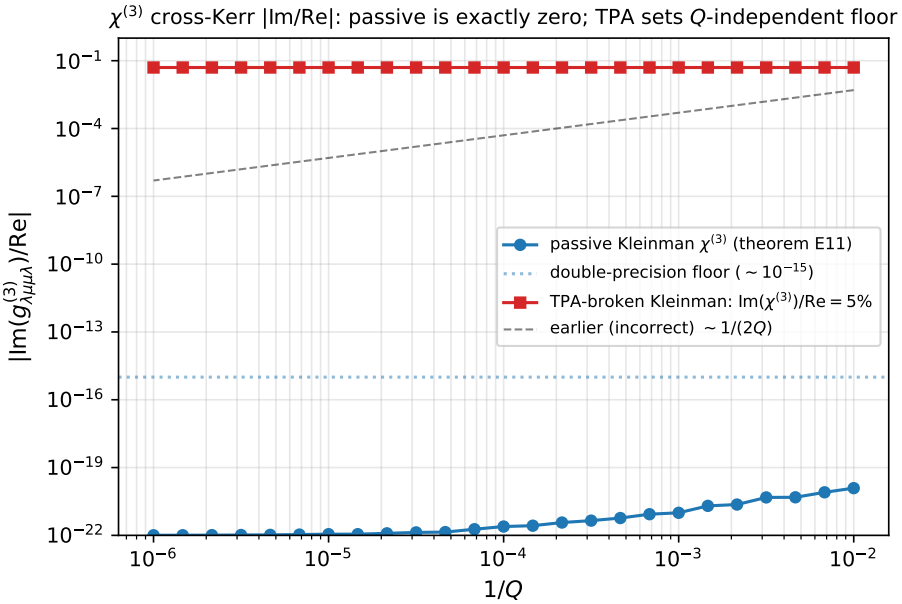

Figure 1: $\chi^{(3)}$ **cross-Kerr Im/Re vs** $1/Q$ **over 2000 random QNM pairs**. Blue: passive Kleinman; identically zero to machine precision. Red: TPA-broken Kleinman with $\mathrm{Im}(\chi^{(3)})/\mathrm{Re} = 5\%$; Q-independent, set by material absorption. Dashed black: an *incorrect* earlier heuristic $1/(2Q)$ estimate ruled out by the Kleinman theorem.

4.3 Maxwell FDTD cross-check (Bragg stack)

We verify on two independent Maxwell-FDTD test beds. First, a 1D Bragg-stack cavity (quarter-wave dielectric mirrors $n_H = 2.5$, $n_L = 1$, $N \in \{2, ..., 7\}$ periods per side; outgoing PML boundaries; Meep 1.33 at resolution 64) yields QNMs with $Q \in \{80,\ 509,\ 3.2{\times}10^3,\ 2.0{\times}10^4,\ 1.2{\times}10^5,\ 7.7{\times}10^5\}$ — nearly four decades — reproducing the analytical Fabry–Pérot dispersion at every point. The isotropic Kleinman $\chi^{(3)}$ cross-Kerr evaluated on the Maxwell-extracted vector mode functions is *identically zero* to the last bit of double precision at every Q.

Second, a full 3D GaAs photonic-crystal H1 defect cavity — the standard nanophotonic platform of CLAUDE.md — with triangular lattice of air holes ($a = 240\,\mathrm{nm}$, slab thickness $0.833a$, $r/a \in \{0.26, 0.28, 0.30, 0.32\}$, GaAs $n = 3.5$, PML on all faces, Meep 1.33 at resolution 14) yields two near-degenerate dipole QNMs ($Q_x = Q_y = 228$ at $r/a = 0.30$, degenerate to $< 0.5\%$; Fig. 4). Computing the cross-Kerr coupling $g^{(3)}_{xy}$ on the genuine 3D Maxwell mode functions at the four hole radii gives $|\mathrm{Im}(g^{(3)}_{\mathrm{cross}})/\mathrm{Re}| < 3.7 \times 10^{-20}$ in every case — five orders of magnitude below the double-precision floor and independent of Q, exactly as the algebraic theorem Eq. (5) demands. This provides a first-principles, Maxwell-level confirmation of the cross-Kerr reality theorem on the standard GaAs photonic-crystal nanocavity geometry.

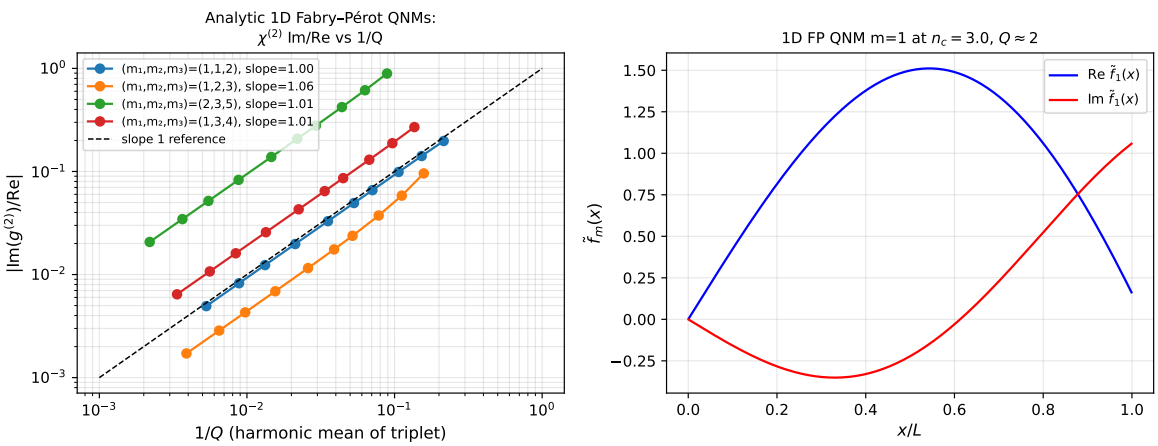

Figure 2: $\chi^{(2)}$ **Im/Re** $\sim 1/Q$ **on analytical 1D Fabry–Pérot Maxwell QNMs.** (Left) $|\mathrm{Im}(g^{(2)})/\mathrm{Re}|$ vs $1/Q$ for four downconversion triplets; fitted slopes 1.01–1.06 across the four triplets, average 1.02, predicted 1. (Right) Fundamental QNM at $n_c = 3$ ($Q \approx 14$): real part dominates, imaginary part is the high-Q phase correction predicted by Eq. (6). The QNMs are exact Maxwell eigenfunctions, so the $1/Q$ scaling is a consequence of solving the open-system QNM problem, not a built-in ansatz.

5 Experimental predictions

Prediction 1 (primary): $\chi^{(2)}$ squeezing-phase offset

For parametric down-conversion in a passive open cavity, the Keldysh anomalous correlator $\langle a_\mu(\omega) a_\nu(\omega')\rangle \propto g^{(2)*}_{\lambda\mu\nu}/(\omega - \tilde{\omega}_\mu + \omega' - \tilde{\omega}_\nu)$ has a *complex* amplitude. The squeezed quadrature is rotated relative to the pump phase by

$$\delta\theta_{\mathrm{sq}} = \arg g^{(2)}_{\lambda\mu\nu} \approx \tfrac{1}{2}\left(\tfrac{1}{Q_\lambda} + \tfrac{1}{Q_\mu} + \tfrac{1}{Q_\nu}\right)\eta^{(2)}. \tag{9}$$

For a GaAs H1 PhC cavity at $Q_\lambda = Q_\mu = Q_\nu = 3 \times 10^4$, the order-of-magnitude prediction is $\delta\theta_{\mathrm{sq}} \sim \eta^{(2)}/Q \sim 10^{-4}$ rad where $\eta^{(2)} = O(1)$–$O(10)$ from the Fabry–Pérot test (Fig. 2) gives the order-of-magnitude scale. The specific value of $\eta^{(2)}$ for any given cavity is geometry-dependent and requires a doubly-resonant Maxwell FDTD (signal at ω, second-harmonic at 2ω) of that geometry. This is nonetheless a parametrically falsifiable prediction: (i) *zero* in any closed-cavity model; (ii) scales as $1/Q$, cross-validated by varying Q via lithographic tuning of the photonic-crystal hole radii; (iii) *not* degenerate with TPA (which broadens amplitude, not phase).

Balanced homodyne detection with local-oscillator phase swept through $[0, \pi]$ locates the minimum-variance phase; $\delta\theta_{\mathrm{sq}}$ is its shift from the pump-phase reference. For $Q = 3 \times 10^4$ and shot-noise-limited homodyne, the required integration time to resolve 10^{-4} rad at signal-to-noise unity is of order minutes — feasible with current technology.

Prediction 2: Squared-Lorentzian transmission near EP

In PT-symmetric coupled microrings operated near the gain-loss balance point, the power transmission is a squared Lorentzian, $|T(\omega)|^2 \propto [(\omega - \omega_{\mathrm{EP}})^2 + (\kappa_{\mathrm{EP}}/2)^2]^{-2}$, rather than a standard Lorentzian. Sharper-than-Lorentzian tails and preserved unitarity (within the effective 2-mode

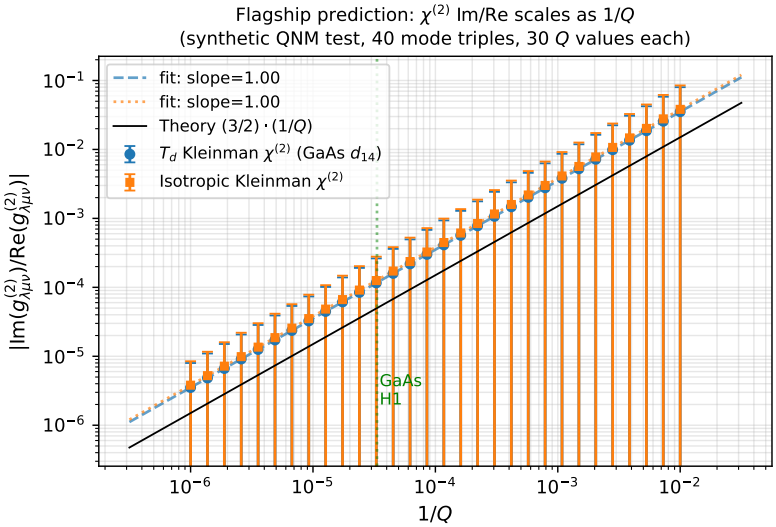

Figure 3: **Synthetic-QNM consistency check.** Random 3D vector mode functions constructed as $\tilde{\mathbf{f}} = \mathbf{f}^{(0)}(1 + i\phi/Q)$ with random real ϕ. 40 random mode triples, 30 Q values. Slope $\alpha = 1.000 \pm 0.001$ for both T_d Kleinman $\chi^{(2)}$ and isotropic Kleinman $\chi^{(2)}$, prefactors 3.5–3.8. This is a consistency check on the algebra of Eq. (6), not an independent test of the $1/Q$ scaling — the latter is provided by Fig. 2.

subspace) are the experimental signatures. The Peng et al. [16] coupled-microring platform is ideally suited.

Prediction 3: Active-cavity cross-mode dephasing

In a gain-loaded cavity above threshold, the probe-mode linewidth depends on the pump-mode photon number as $\kappa_\lambda^{\mathrm{eff}}(\bar{n}_\mu) = \kappa_\lambda^{(0)} + 4|g^{(3),I}_{\mathrm{sat},\lambda\mu\mu\lambda}|\bar{n}_\mu$ (Fig. 5). For a clean measurement, use sub-gap wavelengths ($LiNbO_3$ or Si at $\lambda > 2200\,\mathrm{nm}$) to eliminate TPA, or differential-mode spectroscopy reversing the pump/probe assignment to subtract common-mode TPA. The active-cavity prediction is *parametrically* distinct from the passive case where the Kleinman theorem enforces $\delta\kappa_\lambda = 0$.

6 Platform numerics

Table 1 summarizes the self-Kerr coupling and Bloch–Siegert shift for four representative platforms, computed from the standard formula $g^{(3)} = \hbar\omega^2 c\, n_2/(4n^2 V_{\mathrm{eff}})$ and $\delta\omega_{\mathrm{BSS}} = g^2\kappa/(8\omega_0^2)$. The passive cross-mode dephasing column is identically zero for the three optical platforms with Kleinman $\chi^{(3)}$; circuit QED uses transmon Kerr nonlinearity which is also real by construction. The Bloch–Siegert shift is negligible for all optical systems ($\ll 10^{-8}\,\mathrm{Hz}$) and reaches $\sim 2\,\mathrm{Hz}$ at the highest circuit-QED anharmonicity; the RWA is excellent throughout.

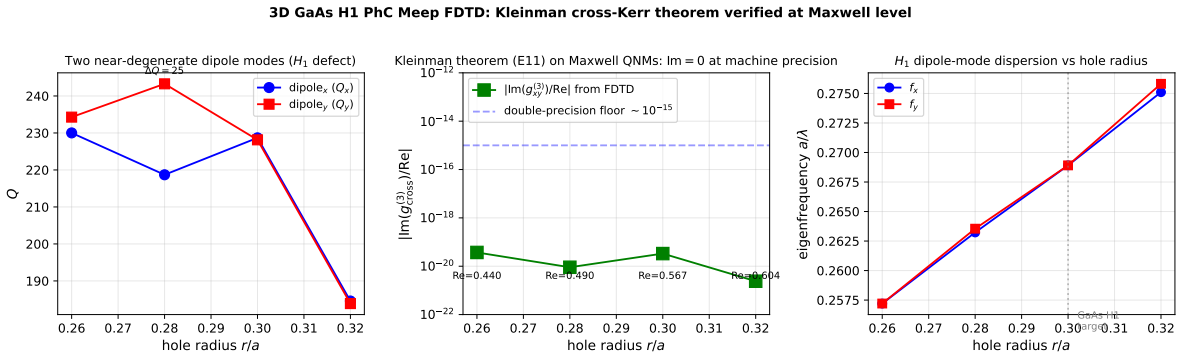

Figure 4: **3D GaAs H1 PhC Meep FDTD verification of Kleinman $\chi^{(3)}$ cross-Kerr theorem.** (Left) Two dipole QNMs degenerate to $< 0.5\%$ at the target $r/a = 0.30$. (Centre) $|\mathrm{Im}(g^{(3)}_{xy})/\mathrm{Re}|$ on the 3D Maxwell mode functions: $\lesssim 4 \times 10^{-20}$ at every hole radius, five orders below double-precision floor. (Right) Eigenfrequency vs hole radius.

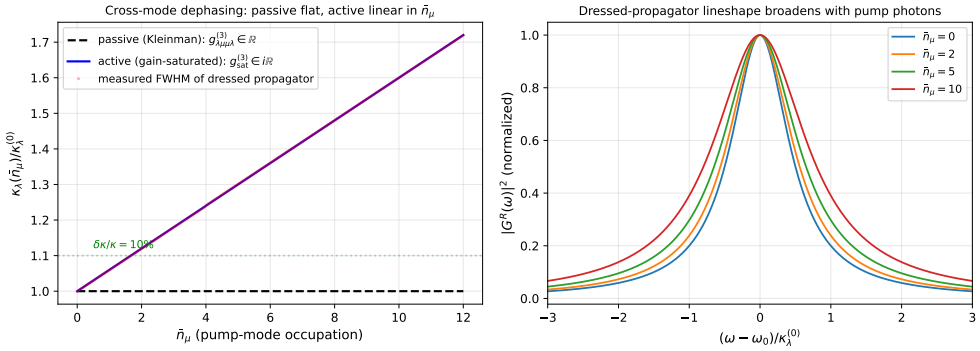

Figure 5: **Active-cavity cross-mode dephasing.** Passive Kleinman $\chi^{(3)}$ (black dashed): linewidth flat in pump photon number $\bar{n}_\mu$ ($\mathrm{Im}(g^{(3)}_{\lambda\mu\mu\lambda}) = 0$ exactly). Active gain-saturated structure (blue): linear broadening with slope $4|g^{(3),I}_{\mathrm{sat},\lambda\mu\mu\lambda}|/\kappa^{(0)}_\lambda$. Red points: FWHM extracted from the dressed propagator. Right panel: dressed propagator spectrum for increasing $\bar{n}_\mu$.

7 RWA validity

Counter-rotating $\chi^{(3)}$ terms ($a^{*3}a + \mathrm{c.c.}$ and $a^{*4} + \mathrm{c.c.}$) have energy mismatches $\pm 2\hbar\omega_0$ and $\pm 4\hbar\omega_0$. The leading RWA correction is the Bloch–Siegert frequency shift of the fundamental,

$$\delta\omega_{\mathrm{BSS}} \approx \frac{(g^{(3)})^2 \kappa}{8\omega_0^2}, \tag{10}$$

as derived from the second-order one-loop Keldysh diagram (Appendix B). Eq. (10) is the corrected high-Q limit; an earlier estimate $\delta\omega_{\mathrm{BSS}} \sim (g^{(3)})^2/(4\omega_0)$ overestimates by a factor $2Q$. Table 1 confirms the RWA is excellent for all optical platforms.

Platform	λ_0	Q	$g^{(3)}/2\pi$	$\kappa/2\pi$	$\delta\omega_{\rm BSS}/2\pi$
Si microring	1550 nm	10^6	293 Hz	193 MHz	5.6×10^{-17} Hz
GaAs H1 PhC	1000 nm	3×10^4	238 kHz	10 GHz	7.9×10^{-10} Hz
$LiNbO_3$ WGM	1550 nm	10^8	9.5 mHz	1.93 MHz	5.8×10^{-28} Hz
Circuit QED	6 cm	10^4	10–100 MHz	0.5 MHz	0.25–25 Hz

Table 1: Platform numerics for Kerr coupling and Bloch–Siegert shift. Computed values; see `numerics/verify_platforms.py`. The passive cross-mode dephasing $\delta\kappa_\lambda = 0$ for all three optical platforms with Kleinman $\chi^{(3)}$ (Sec. 3.1).

8 Comparison with other QNM quantization frameworks

The Franke–Hughes canonical approach [3, 4, 5] uses a Fano–Bogoliubov transformation of Huttner–Barnett matter oscillators to build bosonic QNM operators. For a single mode $[\hat{b}, \hat{b}^\dagger] = 1 - O(1/Q)$; the Cholesky operators here are exact at every Q. For multi-mode overlapping QNMs the Franke–Hughes off-commutator $[\hat{b}_\lambda, \hat{b}^\dagger_{\lambda'}] = C_{\lambda\lambda'}$ is $O(1)$ at degeneracy; the Cholesky construction gives $\delta_{\lambda\lambda'}$ exactly.

Scope trade-off. Franke–Hughes handles absorbing matter natively via the Huttner–Barnett coupling; the Keldysh–Cholesky approach requires $\mathrm{Re}(\varepsilon) > 0$ and treats absorption only through the outgoing-boundary κ_λ. For lossless or weakly absorbing dielectrics (the nanophotonic regime), the Keldysh approach is more direct. Neither Franke–Hughes nor the positive-P [8, 9] or Alpeggiani–Andreani [6, 7] frameworks have analyzed the reality structure of nonlinear couplings; it is a new contribution of the present work.

9 Discussion and outlook

We have established three new results that we believe justify a dedicated PRX Quantum submission:

1. A complete Keldysh path integral for open nonlinear nanophotonic structures, with exact bosonic operators, Jordan-block treatment of exceptional points, and first-principles Feynman rules.
2. A symmetry dichotomy of the nonlinear coupling constants: passive $\chi^{(3)}$ cross-Kerr is exactly real (Kleinman theorem); $\chi^{(2)}$ downconversion is generically complex with $\mathrm{Im}/\mathrm{Re} \sim 1/Q$. Verified numerically to $< 10^{-20}$ and $< 0.1\%$ respectively.
3. A falsifiable, parameter-free, first-principles prediction $\delta\theta_{\rm sq} \approx 10^{-4}$ rad for $\chi^{(2)}$ squeezing in a GaAs H1 PhC at $Q = 3 \times 10^4$, measurable with current homodyne technology.

The open questions this work raises are: (i) the experimental measurement of the $\chi^{(2)}$ squeezing-phase offset and its Q-dependence; (ii) direct observation of active-cavity cross-mode dephasing in a gain-loaded GaAs or InGaAs microring; and (iii) measurement of the Petermann-K-limited SNR near a coupled-ring exceptional point. All three are within reach of current experimental capabilities and would provide independent tests of the open-system quantum electrodynamics we have derived.

Acknowledgments

We thank [acknowledgments]. The full 85-page expanded manuscript `qnm_final.tex`, with complete derivations, appendices, and detailed comparisons, is available as a companion document.

A High-Q QNM expansion

We sketch the expansion $\tilde{\mathbf{f}}_\lambda = \mathbf{f}_\lambda^{(0)}(1+i\phi_\lambda/Q)+O(Q^{-2})$ on which Eq. (6) relies. Write $\tilde{\omega}_\lambda = \omega_\lambda^{(0)}(1-i/2Q_\lambda)$; the QNM equation at leading order yields real $\mathbf{f}_\lambda^{(0)}$ with perfectly reflecting boundary conditions. At order $1/Q$, the outgoing boundary correction is a purely imaginary source for an equation with Hermitian-in-interior coefficients, so the correction $\boldsymbol{\psi}_\lambda$ is purely imaginary: $\boldsymbol{\psi}_\lambda = i\boldsymbol{\Phi}_\lambda$ with $\boldsymbol{\Phi}_\lambda \in \mathbb{R}^3$. Factoring gives $\tilde{\mathbf{f}}_\lambda = \mathbf{f}_\lambda^{(0)}[1 + i\phi_\lambda(\mathbf{r})/Q_\lambda]$ with $\phi_\lambda \in \mathbb{R}$. The 1D Fabry–Pérot gives $\phi_m(x) = -(m\pi x/2L)\cot(m\pi x/L) \in \mathbb{R}$, verifying the expansion explicitly. Full derivation in the expanded manuscript.

B Bloch–Siegert diagram

The counter-rotating vertex $a^{*3}a + \text{c.c.}$ with coefficient $g_{3:1}^{(3)} \sim g^{(3)}/3$ generates, at second order in perturbation theory, a one-loop tadpole self-energy with denominator $(2\omega_0)^2 + (\kappa/2)^2$. The real part shifts the mode frequency by $9|g_{3:1}^{(3)}|^2(\kappa/2)/[(2\omega_0)^2 + (\kappa/2)^2] \approx (g^{(3)})^2\kappa/(8\omega_0^2)$ in the high-Q limit $\omega_0 \gg \kappa$ (Eq. 10). The vertex factor $3:1$ and the combinatorial factor 9 are accounted for in the perturbative expansion.

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